

DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES E INFORMÁTICA

MESTRADO EM ENGENHARIA DE COMPUTADORES E TELEMÁTICA

ANO 2026/2027

MODELAÇÃO E DESEMPENHO DE REDES E SERVIÇOS

MINI-PROJECT 1:

PERFORMANCE EVALUATION OF POINT-TO-POINT LINKS SUPPORTING MULTIPLE PACKET FLOWS

Mini-Project 1

Assignment Rules

- Develop the mini-project in a group of 2 students.
- Implement all tasks using MATLAB and write a report containing all results together with their analysis and conclusions as complete as possible.
- When requested, include in the report the developed MATLAB codes duly explained.
- The report must identify the elements of the group (names and student numbers) and must include in the first page an auto-evaluation of the work percentage done by each element.
- In the classes of 11 of November (TP1) and 12 of November (TP2), both elements of each group must attend the class and demonstrate what have already developed.
- The report must be sent in PDF format to asou@ua.pt **until 23:59 of 17th of November**.

Introduction

The aim is to address the performance evaluation of a link of $C = 10$ Mbps supporting 3 flows, whose packet statistics are:

- Flow 1 with a random packet size between 100 and 1500 bytes (all values with equal probability)
- Flow 2 with a random packet size between 100 and 150 bytes (all values with equal probability)
- Flow 3 with a random packet size between 500 and 1000 bytes (all values with equal probability)

In all Tasks, consider the 4 following traffic cases for the 3 flows:

Case A: the packet arrival rates are $\lambda_1 = 400$ pps, $\lambda_2 = 400$ pps and $\lambda_3 = 600$ pps,

Case B: the packet arrival rates are $\lambda_1 = 600$ pps, $\lambda_2 = 600$ pps and $\lambda_3 = 600$ pps,

Case C: the packet arrival rates are $\lambda_1 = 800$ pps, $\lambda_2 = 800$ pps and $\lambda_3 = 600$ pps.

Case D: the packet arrival rates are $\lambda_1 = 1000$ pps, $\lambda_2 = 1000$ pps and $\lambda_3 = 1000$ pps.

In all experiments requesting simulation results, present always the estimated values and 90% confidence intervals based in 50 runs of the requested simulator with a stopping criterion of $P = 100.000$ ($= 10^5$) each run. When requested to plot the results, present the estimated values in bar charts and the 90% confidence intervals in error bars on the same plot.

Task 1

1.a. **(Evaluation weight: 10%)** Develop a new version of the simulator *Sim2Flows* to support the 3 flows specified in the Introduction. Include in the report the developed MATLAB function of the new simulator version highlighting and justifying the changes introduced.

1.b. **(Evaluation weight: 5%)** Consider the queue size $f = 1.000.000$ Bytes. Develop a MATLAB script to estimate by simulation all performance parameters of each flow for case C. Present and justify the observed results between the different flows.

1.c. **(Evaluation weight: 5%)** Consider case C modelled by an $M/G/1$ model. Develop a MATLAB script to determine the theoretical values of the average packet delay and throughput of each flow. Include in the report the developed MATLAB code duly explained. Present the results, compare them with the simulation results of **1.b** and draw all relevant conclusions.

1.d. **(Evaluation weight: 5%)** Consider again the queue size $f = 1.000.000$ Bytes. Develop a MATLAB script to estimate by simulation the average packet delay and the maximum

packet delay of Flow 3 for the cases A, B and C. Plot the simulation results. Justify the results and draw all relevant conclusions.

Task 2

2.a. (Evaluation weight: 10%) Develop a new version of simulator *Sim2FlowsPrio* (developed in Practical Assignment no. 2) to support the 3 flows specified in the Introduction and consider that Flow 3 has the highest priority and Flows 1 and 2 have the second highest priority. Include in the report the developed MATLAB function of the new simulator version highlighting and justifying the changes introduced.

2.b. (Evaluation weight: 10%) Consider the queue sizes $f_1 = f_2 = 1.000.000$ Bytes. Develop a MATLAB script to estimate by simulation all performance parameters of each flow for case C. Present and justify the observed results between the different flows. Justify also the differences between these results and the ones in **1.b**.

2.c. (Evaluation weight: 5%) Consider case C modelled by an $M/G/1$ model with priorities. Develop a MATLAB script to determine the theoretical values of the average packet delay of each flow. Include in the report the developed MATLAB code duly explained. Present the results, compare them with the simulation results of **2.b** and draw all relevant conclusions.

2.d. (Evaluation weight: 10%) Consider again the queue sizes $f_1 = f_2 = 1.000.000$ Bytes. Develop a MATLAB script to estimate by simulation the average packet delay and the maximum packet delay of Flow 3 for the cases A, B and C. Plot the simulation results. Justify the results, compare with the results of **1.d** and draw all relevant conclusions.

Task 3

3.a. (Evaluation weight: 10%) Develop a new version of the simulator *Sim2FlowsDRR* (developed in Practical Assignment no. 2) to support the 3 flows specified in the Introduction and consider that each round-robin round is by the order of the flow ID. Include in the report the developed MATLAB function of the new simulator version highlighting and justifying the changes introduced.

3.b. (Evaluation weight: 10%) Consider the queue sizes $f_1 = f_2 = f_3 = 1.000.000$ Bytes. Develop a MATLAB script to estimate by simulation all performance parameters of each flow for case C considering a Quantum value of 1500 for each flow. Present and justify the observed results between the different flows. Justify also the differences between these results and the ones in **1.b** and **2.b**.

3.c. (Evaluation weight: 5%) Repeat experiment **3.b** but now for the congested case D. Present and justify the observed results between the different flows. Justify also the differences between these results and the ones in **3.b**.

3.d. (Evaluation weight: 5%) Repeat experiment **3.c** for the congested case D but now considering a Quantum value of 1500 for Flow, 1500 for Flow 2 and 4500 for Flow 3. Present and justify the observed results between the different flows. Justify also the differences between these results and the ones in **3.c**.

3.e. (Evaluation weight: 10%) Compute theoretically the ideal throughput of each flow (following the weighted max-min fairness principle) for the cases addressed in **3.c** and **3.d**. Present the ideal throughput values together with the calculations, compare them with the simulation results of **3.c** and **3.d** and draw all relevant conclusions.